Student Project: Hazard Identification Checklist

Project Title: Plant-Saver

Project Advisor

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Project Safety Coordinator:

Name:	Department	phone	e-mail
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Project Location(s) (Department, building, room#)

DOW 901

Summary of Project Objectives:

Create a device to take measurements around the care of a plant to suggest changes to the environment.

Hazard Checklist: Check all that apply to your project

- ☐ Biological: Recombinant DNA; infectious or potentially infectious materials (bacteria, viruses) ; animals and insects (bites)
- ☐ Chemical: Laboratory chemicals, gasoline, solvents, lubricants, paints, etc.
- ☒ Electrical: Batteries; Control Panels; High Voltage > 50V; High Current.
- ☐ Hazardous procedures: Metal Fabrication (Welding, Cutting, Brazing, Grinding, Drilling, Machining, etc.) Soldering, Construction, etc.
- ☐ Mechanical: Power Tools and Equipment; Machine Guarding / Power Transmission (Gears, Belt/Chain drives, Rotating/ Moving parts, Pinch points), Robotics, Sharp objects, Stored energy (springs, gravity, etc).
- ☐ Noise: Greater than 90 decibels (heavy traffic, power lawn mower, screaming child)
- ☐ Physical: Extreme temperatures; Lifting heavy objects; Heights (scaffolding, ladders, cliffs, trees); Slip, trip or fall hazards; water; Overhead/Falling objects (cranes, hoists, tree branches, drones, projectiles); Vehicle traffic; Confined spaces; Dust; etc.
- ☐ Pressure: Hydraulics, Pneumatics, Fuel injectors, Pressurized fluids, Compressed air / gases, Steam
- ☐ Radiation: Non-ionizing (Laser, Microwave, UV, Infrared light, etc.,) and Ionizing (Radioactive isotopes, X-rays, etc.,)
- ☐ Other Hazards not specified above:
- ☐ There are NO Hazards associate with this project.

Electrical Safety

☒ Batteries

Batteries are generally a safe and reliable source of electrical energy. However, improper use of a battery may cause injuries. For example, electrical shock from a short circuit, physical burns due to arcing or fires may occur if a battery is not properly used or stored. Similarly, both thermal and chemical burns may result from overheating and rupture if batteries are not recharged correctly. The type of battery, its voltage and the current it is able to generate should all be taken into consideration when evaluating the risks associated with the use of a battery. Additionally, the weight of lead/acid batteries should be considered a potential hazard when these batteries must be moved. Precautions should be taken to prevent lifting injuries. Finally, many of the chemicals used in batteries are toxic or hazardous to the environment. Consideration should be give when these batteries are disposed.

1. Will the batteries be recharged? Yes
2. Does the battery store sufficient energy to cause arcing or electrical shock? No

Note: Arcing / short circuit of 9-volt batteries can cause fire. see 9-volt battery safety

3. Describe precautions that will be used to prevent injury when lifting / handling heavy batteries

The batteries in this project are not heavy. There are no special precautions needed.

4. What is the proper method for disposal of batteries used in this project?

There are no plans to dispose of the batteries used in this project, but in that case where that would be necessary they should be disposed of via a dedicated battery disposal services.

☐ High Voltage (Energized- Exceeding 50V)

☐ High Voltage (De-Energized-Exceeding 50V)

☐ Low Voltage, High Current (batteries, power supplies)

☒ Wet Conditions

1. Describe the electrical hazards (objects, materials, etc.) that are associated with your work.

We have a battery and a PCB.

2. What are the risks associated with these objects and/or the working environment?

Water contact may short leads on the PCB or battery causing the device to fail.

3. Describe the engineering controls, work practices, training and personal protective equipment that will be used to minimize these risks.

Engineering Controls:	The housing design will be engineered to prevent excess water from reaching sensitive components.
Work Practices:	Water will not be applied directly to the housing.
Training:	No extra training is required.
Personal Protective Equipment (PPE):	No PPE is necessary.

4. Provide a brief summary of your safe operating procedure.

Before applying power to the device, ensure that all seals are intact, fittings are secure, and that the housing has no large defects which would allow water ingress to occur.

Review and Approval

Name of Project Advisor	Signature and Date
Name of Project Safety Coordinator	Signature and Date

Project Team Members

I have received all the safety training related to this project and have read the safety manuals and safety data sheets (SDS) relevant to the project. I understand and will abide by all required safety precautions while working on this project, including but not limited to following safe operating procedures and wearing appropriate personal protective equipment.

	Name of Project team member	Signature and Date
Remove this team member	Jackson Wiles	Jackson Wiles -03/10/2026
Remove this team member	Sebastian Stauber	Sebastian Stauber-03/10/20206
Remove this team member	Donovan Ghysels	Doinovan Ghysels - 03/10/2026
Remove this team member	Noah Schmidt	Noah Schmidt - 03/10/2026

Remove this team member	Brandon Early	Brandon Early -03/10/2026
	Name of Project team member	Signature and Date

Add Another Team member

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